Roll. No._____

THAPAR UNIVERSITY, PATIALA

SCHOOL OF ENERGY AND ENVIRONMENT

BE/B.Tech II Year **UEN002-Energy and Environment**
End-Semester Exam - May 2017 **Duration : 3 h**
Instructor
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Please note the marks assigned for each of the questions/sub-questions and answer accordingly
The answers should strictly follow the serial order.

1. (a) As India is facing an increasing demand-supply gap in energy, it is important to harvest the solar potential to meet the energy needs. In this context, analyze the availability of solar energy as a resource in India. Formulate various strategies to tap this resource. (20) (10+5+5)
   (b) Deforestation influences the climate and contributes to a loss of biodiversity. Discuss the various drivers for deforestation.
   (c) Describe the mechanisms involved in particle collection by fabric collector.
2. (a) Fresh water insecurity has been recognized as one of the biggest crisis facing the world today. Analyze the human contribution the problem of water insecurity. Forecast the implications of an increasingly water insecure world, with suitable examples. (20) (10+5+5)
   (b) Write briefly on the differences between plate and tube type electrostatic precipitators (ESPs).
   (c) Write briefly the principles and advantages of attached growth processes in biological treatment systems.
3. (a) Food and Agricultural Organization (FAO) of the United Nations has identified population growth and depleting resources as the major challenges in achieving food security in the 21st century. In this context, review the extent of the challenge posed by depleting resources and growing population to food security. (20) (10+5+5)
   (b) What do the three arrows of the recycling symbol depict? What are the work elements that would comprise integrated waste management system?
   (c) The utilization of mineral resources has been showing a dramatic increase in the past century. This exploitation of mineral resources is associated with various environmental impacts. Briefly evaluate the environmental impacts of mining.
4. (a) The waste collected from a particular locality has following composition: (20) (10+5+5)

Component	% by wt	Un-compacted Bulk Density (kg/m^3)
Corrugated Cardboard	25	30
Paper products	15	61
Aluminum	9	38
Food waste	29	368
Yard waste	22	7.1

1) What is the bulk density for the waste mixture prior to the compaction? Assume that the compaction in the landfill cell is 500 kg/m^3.
2) Estimate the volume reduction (expressed as %) during compaction in the landfill.
3) If food and yard waste are diverted for composting, what is the un-compacted bulk density of the remaining waste?

   (b) Differentiate between coagulation and flocculation? Outline the parameters on which separation of suspended solids in waste water depend?
   (c) Describe any two physical and two chemical properties of municipal solid waste and their importance in waste management perspective.

5. (a) Chernobyl nuclear disaster has been classified on the scale of 7 on the international scale of nuclear disasters. Identify the causes leading to the Chernobyl nuclear disaster. Analyze the health and environmental consequences of this disaster. (20) (10+5+5)

(b) A wastewater stream has a BOD of 4000 mg/L. Calculate the flow rate of BOD (in kg/hr) if the wastewater has a flow rate of 5000 m^3/hr. Assume, if necessary, the density of wastewater as 1.05 gm/cm^3

(c) The Northern part of India has major issue of stubble burning and associated air pollution. One of the important options for utilization of the stubble residue is Biomass gasification. What are the advantages and limitations of Biomass Gasification as an energy source?

Best Wishes….